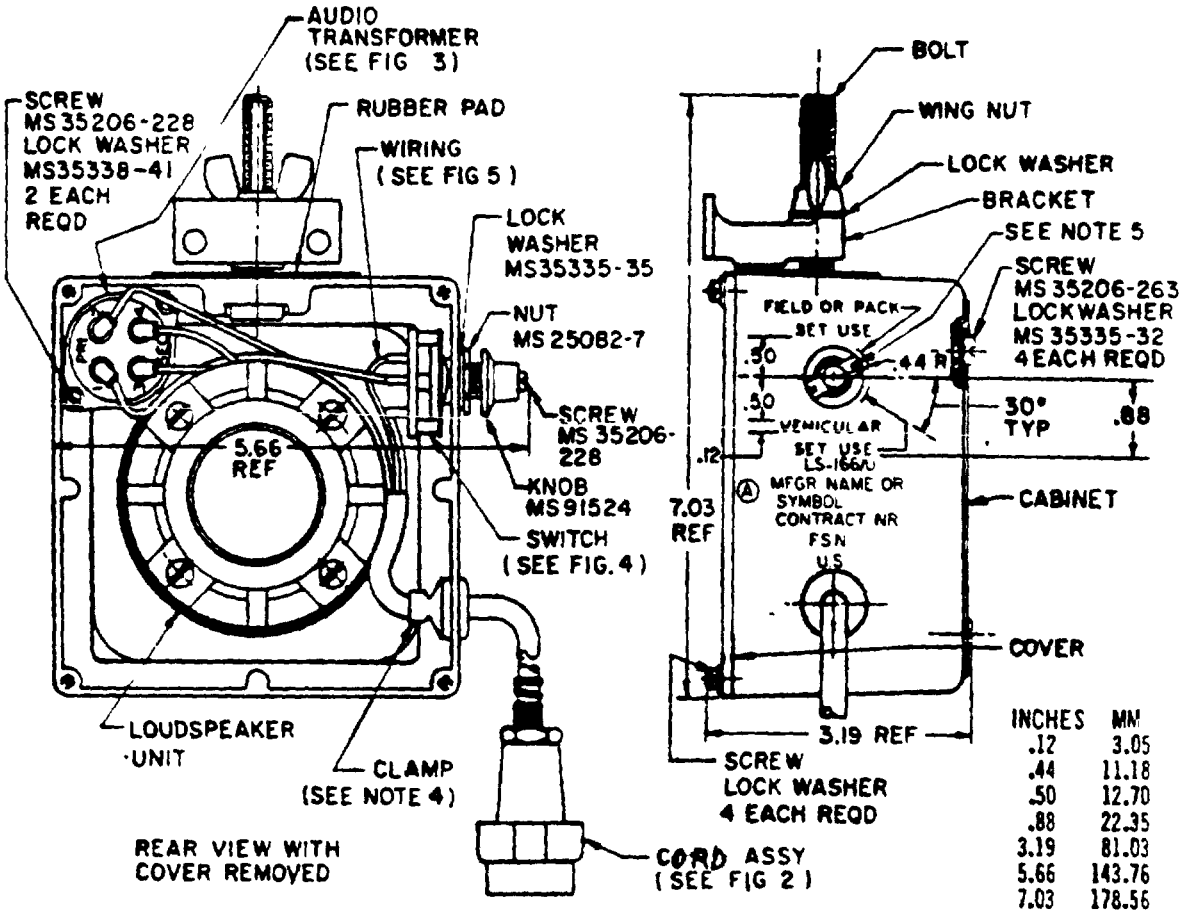
MILITARY SPECIFICATION SHEET

LOUDSPEAKER, PERMANENT MAGNET (ENCASED, 2-WATT, FUNGUS-, GUNBLAST-, AND IMMERSION-RESISTANT), TYPE LS-166/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the loudspeaker described herein shall consist of this document and the latest issue of Specification MIL-L-12632



INCHES	MM
.12	3.05
.44	11.18
.50	12.70
.88	22.35
3.19	81.03
5.66	143.76
7.03	178.56

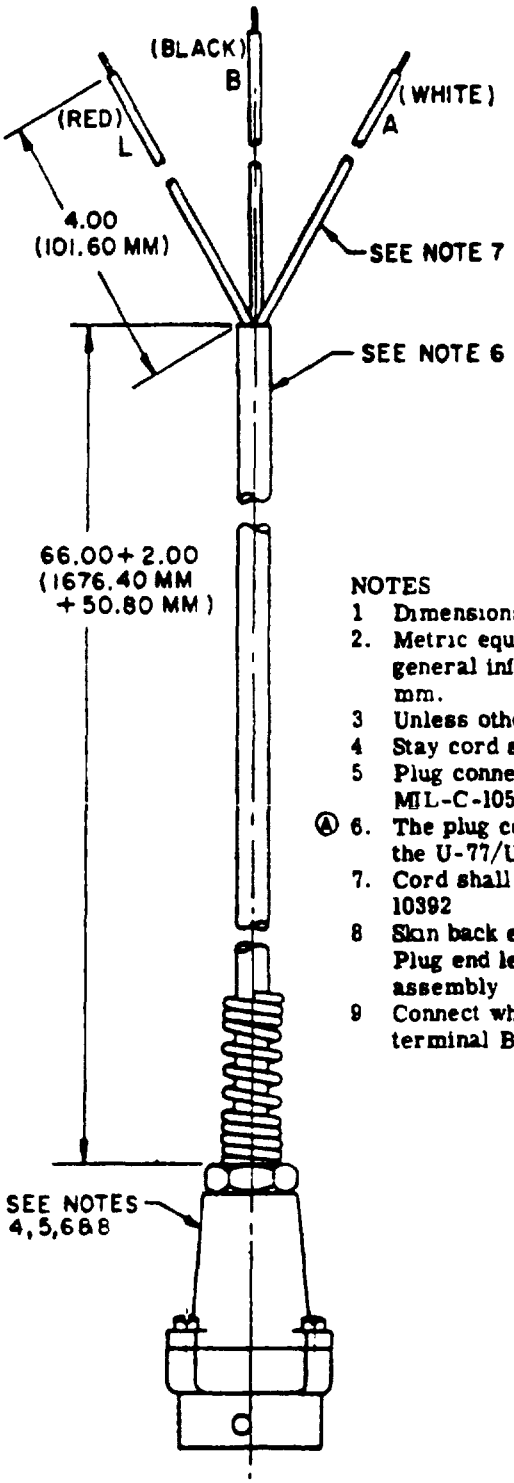
NOTES:
1. Dimensions are in inches
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25 4 mm
3. Unless otherwise specified, tolerance is ± .02 (.51 mm) and ± 1/2° on angles.
4. Steel clamp firmly clamped on cable jacket. Design optional A strain relief bushing (e g Heyco) is a suitable substitute for the grommet/clamp combination
5. Drill 391 (9 93 mm) diameter hole for rotary switch.

FIGURE 1. Loudspeaker LS-166/U.

Ⓐ denotes changes.

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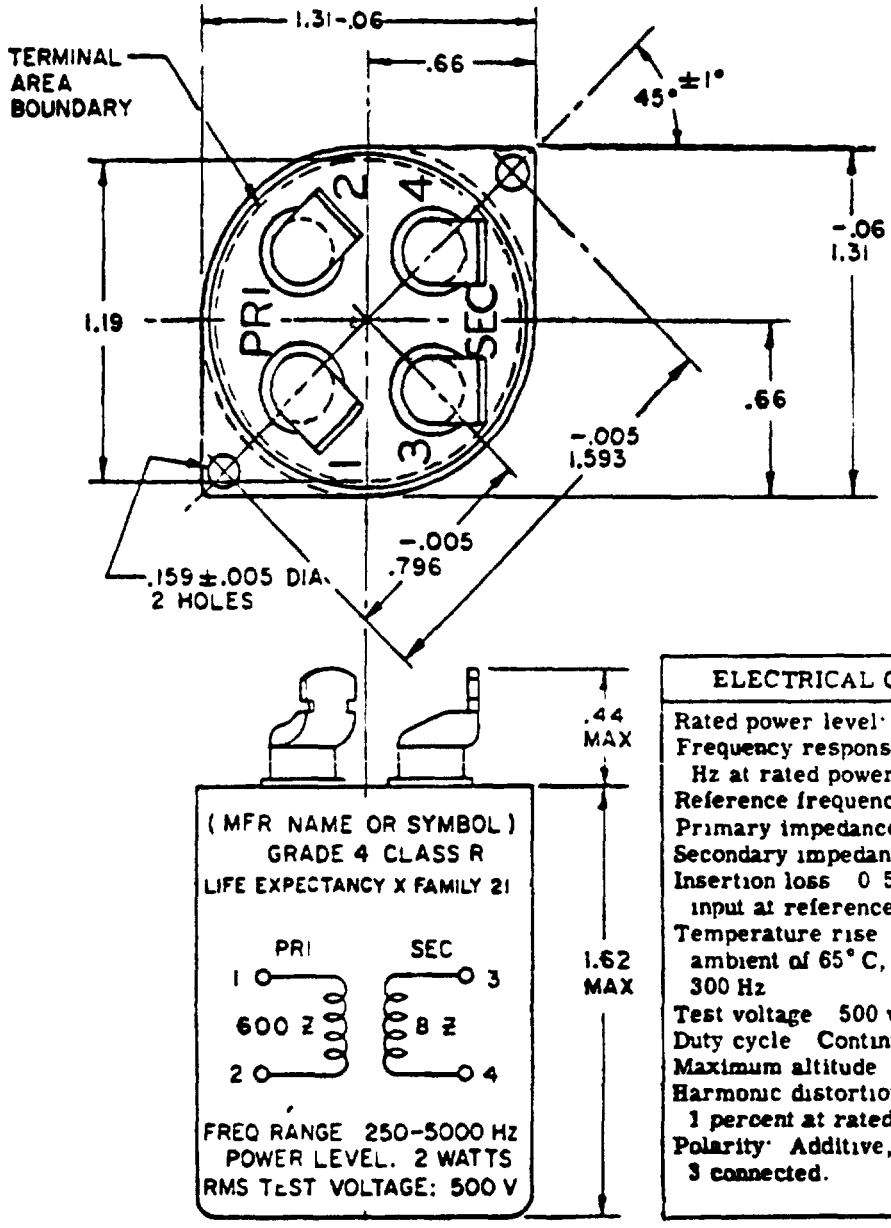
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NOTES
1. Dimensions are in inches
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25 4 mm.
3. Unless otherwise specified, tolerance is ± 06 (1 52 mm).
4. Stay cord shall be tied securely to stay ring of plug.
5. Plug connector shall be type U-77/U in accordance with MIL-C-10544.
Ⓐ 6. The plug connector type U-161/U is a suitable substitute for the U-77/U
7. Cord shall be type WT-15A/U in accordance with MIL-C-10392
8. Skin back ends .12 (3.05 mm), twist and solder coat. Plug end leads shall be 1.12 (28.45 mm) long before assembly
9. Connect white conductor to terminal A, black conductor to terminal B, and red conductor to terminal L of plug U-77/U

FIGURE 2 Cord assembly.

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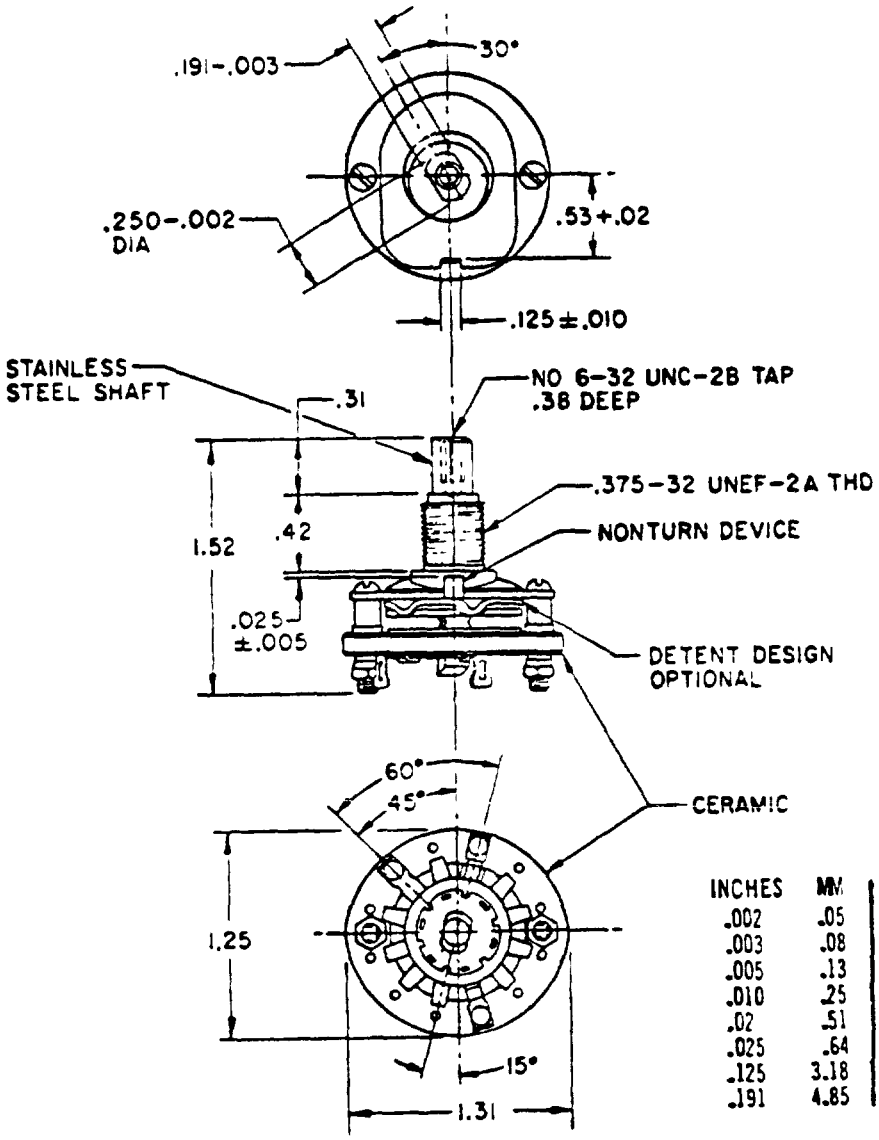
INCHES	MM
.005	.13
.06	1.52
.159	4.04
.38	9.65
.44	11.18
.66	16.76
.796	20.22
1.19	30.23
1.31	33.27
1.593	40.46
1.62	41.15

ELECTRICAL CHARACTERISTICS
Rated power level: 2 watts, ±1 db
Frequency response range 250 to 5000 Hz at rated power, flat within 2 db.
Reference frequency 1000 Hz
Primary impedance 600 ± 60 Ω.
Secondary impedance 8 Ω.
Insertion loss 0 5 db, max, with 2 watts input at reference frequency
Temperature rise 40° C, max, above an ambient of 65° C, with rated power at 300 Hz
Test voltage 500 volts, rms
Duty cycle Continuous
Maximum altitude 10,000 feet
Harmonic distortion: Not more than 1 percent at rated power.
Polarity: Additive, with terminals 2 and 3 connected.

NOTES:
1. Dimensions are in inches
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.02 (.51 mm).
4. Solder lug terminals, No. 18 AWG
5. Markings shall be as shown

FIGURE 3. Audio transformer.

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INCHES	MM	INCHES	MM
.002	.05	.250	6.35
.003	.08	.31	7.87
.005	.13	.38	9.65
.010	.25	.42	10.67
.02	.51	.53	13.46
.025	.64	1.25	31.75
.125	3.18	1.31	33.27
.191	4.85	1.52	38.61

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25 4 mm
3. Unless otherwise specified, tolerances are ±.02 (51 mm) and ±1/2°.
4. Construction - single section, single pole
5. Contacts shall be non-shorting
6. Stops shall be provided to limit rotation to two positions.
7. Terminals - angle of bend 60° to 90°.
8. Torque (rotational) At 25° +10° -5°C: 1.5 pound-inches min: 6 pound inches max.
   At -55° ±2°C: 1.5 pound-inches min: 8 pound inches max.
9. Life, rotational 10,000 rotational cycles at +125°C.

FIGURE 4. Rotary switch.

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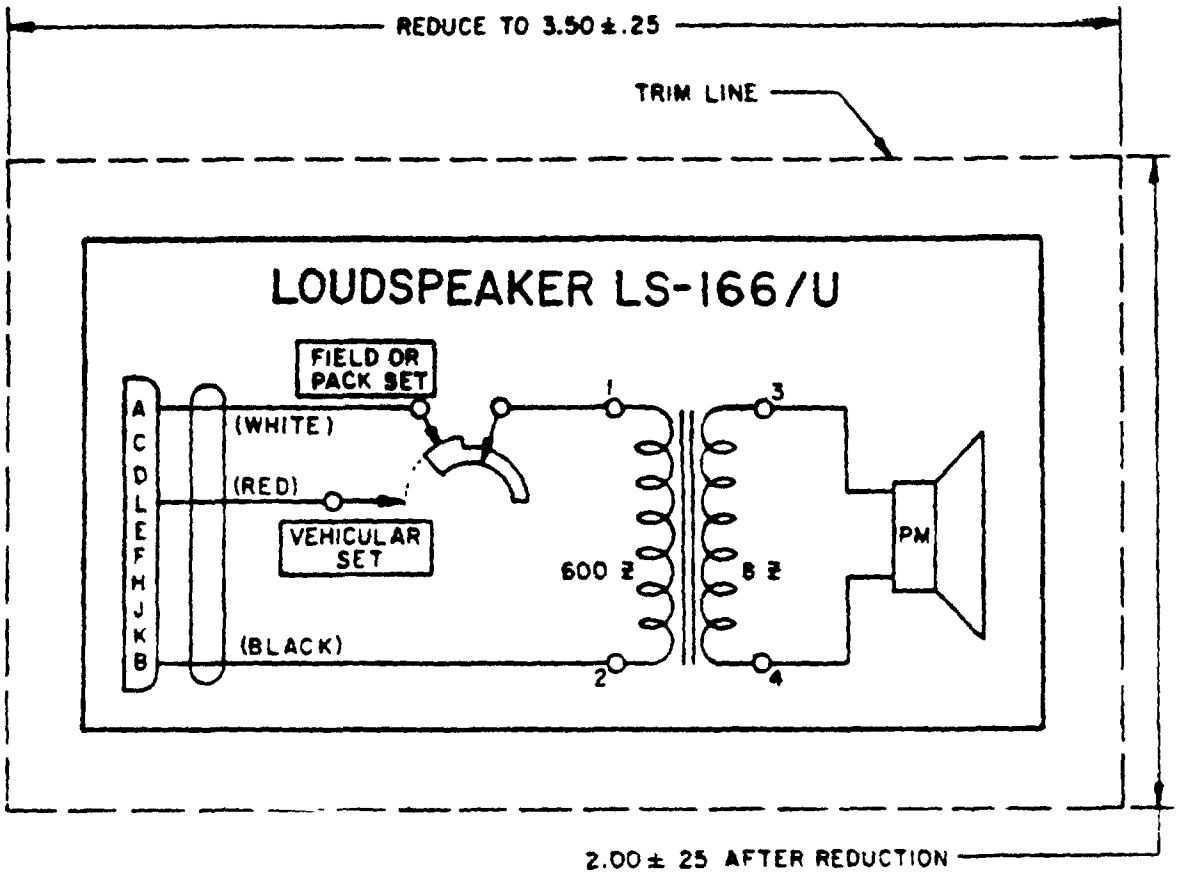


FIGURE 5. Schematic diagram and label size

REQUIREMENTS:

Design and construction (see figure 1)
Loudspeaker unit - Type LS-445/U of MIL-L-12606.
Cabinet - In accordance with MIL-L-12632.
Cord assembly - See figure 2.
Cable - Type WT-15A/U of MIL-C-10392.
Plug - Type U-77/U of MIL-C-10544.
Audio transformer - See figure 3.
Rotary switch - See figure 4.
Wiring - See figure 5.

Schematic diagram label: In accordance with MIL-L-12632 and figure 5.

Frequency response test voltage (rms). 36.0 volts.

Marking See figure 1

Custodians
Army - EL
Navy - EC
Air Force - 85

Review activities
Army - EL
Navy -
Air Force - 80, 13
DSA - ES

User activities
Army - ME
Navy - MC, CG, OS
Air Force -

Preparing activity
Army - EL

Agent:
DSA - ES

(Project 5965-0141-2)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION MIL-L-12632/2A LOUDSPEAKER, PERMANENT MAGNET (ENCASED, 2-WATT, FUNGUS-, GENELAST-, AND IMMERSION-RESISTANT), TYPE LS-166/U

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

ESC-PM 1068-68

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